

Module 6 Worksheet

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Worksheet Information

Goal of the worksheet:

The Module 6 worksheet is an opportunity to apply decomposition-based tests to data and to better understand their reliance on the assumptions of linear regression models. By completing this worksheet, each student will be developing the skills to achieve the following weekly learning objectives:

- State appropriate hypotheses for each decomposition-based test and correctly conduct and conclude each test in context.
- Explain the limitations of the hypothesis tests arising from violated assumptions.
- Integrate decomposition-based hypothesis tests into decision process when building models.

This worksheet, in addition to the remainder of the class time, is **important practice for completing questions on the final exam, and for your final project.**

Preparation assumed:

- Please complete this worksheet after attending your in-person lectures for each week. If you did not attend class, please review the annotated slides posted on Quercus.
- Additional R/Coding resources are also linked on the Quercus homepage.

How to complete this worksheet:

- Students may work and discuss this worksheet in small groups of 2-3 if desired. However **each student** must submit their worksheet to MarkUs to receive their completion credit. Each student must work on their **own copy** of the assignment.
- All the code and course knowledge needed to complete this worksheet has been provided in the class materials. It may help to have these open while working on this document.
- Follow the instructions provided in each question to complete the code.
- DO NOT change the names of the variables that store your final answers.
- When in doubt about a question in the worksheet or your code, ask a TA or the instructor during office hours or on the discussion board.

Steps for submitting to MarkUs:

1. Go to [MarkUs](#) and log in using your UofT credentials.
2. Select the STA302 course from the list.
3. Select Worksheet 6 from the assignment list.
4. Under Submissions, upload your Rmd file and select the file name from the list.

5. Go to Automated Testing and select Run Tests to check your worksheet answers.
6. You can submit as many times as you want, but only your latest submission before the deadline will be counted.
7. It is recommended you submit your file to [MarkUs](#) after you complete each activity to check your answers before moving on. You can submit multiple times to check your work, as your autograding tokens regenerate over time.
8. Even if you are out of autograding tokens, you can still submit your work. You just cannot check if your answers are correct yourself, but the tests will still be run by the instructor for grading.
9. If you are having trouble with the submission process, please post on the discussion board.

What to do if a test fails on MarkUs

1. Don't panic. Your work won't really be graded until the deadline, so start early to make sure you have lots of time to resolve issues before the deadline.
2. Read the message to get hints about what the problem is. For example "variable X not present" means that you may have a typo in your variable name.
3. Search on the discussion board to see if other classmates have encountered a similar error (and if not, consider posting a screenshot of the error message).
4. Sometimes, it will resolve the problem if you type out your values as a number (e.g. `var <- 1`) instead of a function (e.g. `var <- round(...)`).
5. Come to TA or instructor office hours with your issue.

This week, we will be working with a dataset that comes with lots of interesting problems. The data `pgatour2006.csv` has been loaded to JupyterHub with your RMD worksheet. It contains the variables:

- Y , PrizeMoney = average prize money per tournament
- X_1 , Driving Accuracy is the percent of time a player is able to hit the fairway with his tee shot.
- X_2 , GIR, Greens in Regulation, is the percent of time a player was able to hit the green in regulation. A green is considered hit in regulation if any part of the ball is touching the putting surface and the number of strokes taken is two or less than par.
- X_3 , Putting Average measures putting performance on those holes where the green is hit in regulation (GIR).
- X_4 , Birdie Conversion % is the percent of time a player makes a birdie or better after hitting the green in regulation
- X_5 , SandSaves % is the percent of time a player was able to get “up and down” once in a greenside sand bunker.
- X_6 , Scrambling % is the percent of time that a player misses the green in regulation, but still makes par or better.
- X_7 , PuttsPerRound is the average total number of putts per round.

We will be using this dataset to practice conducting ANOVA and partial F tests and documenting our decisions along the way.

1) Fitting a Full Model with 7 Predictors

Seven predictors are listed in the description above. Begin by loading the dataset `pgatour2006.csv` and looking at the `summary()` of this dataset. You are looking for the names of these 7 predictor variables and the response.

Fit a multiple linear model using Prize Money as the response and the 7 other variables above as predictors. Replace NULL in the variable `full1` with this linear model fit using `lm()`.

Display the full model summary of this `full1` using `summary(full1)`. How many predictors coefficients are not statistically different from 0 in this model? Store this integer value in `count` by replacing NULL with the number.

```
# load the data
pga <- read.csv(file="pgatour2006.csv", header=T)

# fit the linear model with 7 listed predictors and response Prize Money
full1 <- NULL
#summary(full1)

# how many coefficients are not statistically different from 0?
count <- NULL
```

Question: What is the conclusion of the ANOVA overall test of significance for this model?
TYPE YOUR ANSWER BELOW:

2) Fitting a Reduced Model

Fit a new model using the `lm()` function that uses Prize Money as the response and includes only the significant predictors that were present in `full1`. Store this new model in the variable `reduced1` by replacing the NULL with the completed `lm()` model.

Perform a Partial F test to compare the `full1` model with the `reduced1` model using the `anova()` function. Remember that the model with fewer predictors should be included in the function first, followed by the model with more predictors.

Replace NULL in the variable `Fvalue1` with the value corresponding to the F statistic for this test. Enter this value as a number as it appears in the output from `anova()`.

```
# fit reduced model without insignificant predictors
reduced1 <- NULL

# conduct a partial F test

# what is the value of F statistic?
Fvalue1 <- NULL
```

Question: What does this F statistic measure or tell us about the sums of squares of the two models?
TYPE YOUR ANSWER BELOW:

3) Check assumptions of full model

Begin by extracting the residuals and fitted values from the `full1` model to be used to create the plots. Store these in `e_hat` and `y_hat` respectively by replacing `NULL` with the appropriate functions (see previous worksheets or Module 3 code for review).

```
# extract residuals and fitted values from full1
e_hat <- NULL
y_hat <- NULL
```

Run the code below to produce ALL the plots needed to perform a complete residual analysis. You will need to uncomment these lines first. The `for` loop simply runs through all the predictors in the dataset and makes a Residuals vs Predictor plot for each.

```
#pairs(pga[,c(5,6,7,8,9,10,12)])
```

Question: What assumptions may be violated in this model?
TYPE YOUR ANSWER BELOW:

Let's try a transformation on the response variable. In the variable `lnPrizeMoney`, replace `NULL` with a natural logarithm transformation of `pga$PrizeMoney`. Then fit a new full model in `full2` that uses the same predictors as `full1` but with `lnPrizeMoney` as the response.

```
# transform PrizeMoney with log()
lnPrizeMoney <- NULL

# fit a new full model using transformed response
full2 <- NULL
```

Finally, recheck a subset of these residual plots (the QQ plot, residual vs fitted, and condition 1). Start by extracting the residuals and fitted values from `full2` and store these in `e_hat2` and `y_hat2` respectively. Then uncomment the plot code to view the new plots.

```
# extract residuals and fitted values
e_hat2 <- NULL
y_hat2 <- NULL

# uncomment to see new plots
#par(mfrow=c(1,3))
#plot(lnPrizeMoney ~ y_hat2, ylab="Response", xlab="Fitted")
#plot(e_hat2 ~ y_hat2, ylab="Residuals", xlab="Fitted")
#qqnorm(e_hat2)
#qqline(e_hat2)
```

4) Redo reduced model

Look at the summary of the new full model `full2` using the `summary()` function. How many predictors are now not significantly different from 0?

Fit a reduced model by including only the predictors that are significantly linearly related to the response in `full2`. Store this model in `reduced2`. Be sure to use the transformed response `lnPrizeMoney` in this model.

Finally, conduct a Partial F test to see whether we would prefer the full model or the reduced model. Again use the `anova()` function to do this. Replace `NULL` in the variable `Fvalue2` with the value corresponding to the *F statistic* for this test. Enter this value as a number as it appears in the output from `anova()`.

```
# look at summary of full2
```

```
# fit model that removes insignificant predictors  
reduced2 <- NULL
```

```
# conduct partial F test
```

```
# what is the F statistic?  
Fvalue2 <- NULL
```

Question: What is the conclusion of this partial F test?
TYPE YOUR ANSWER BELOW:

END OF WORKSHEET - BE SURE TO SUBMIT YOUR WORKSHEET ON MARKUS TO RECEIVE COMPLETION CREDIT